MIL-P-37018
9 January 1976

MILITARY SPECIFICATION

PAD, COOLING, CHEMICAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a disposable, plastic, chemical ooling pad suitable for providing a therapeutic level of coolness.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in 'fect on date of invitation for bids or request for proposal, form a rt of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| PPP-B-636 | Boxes, Shipping, Fiberboard. |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | Marking for Shipment and Storage. |

(Copies of specifications and standards required by suppliers in onnection with specific procurement functions should be obtained from he procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The chemical cooling pad shall consist of the materials pecified herein. Dimensional tolerance on the thickness of the plastic heet or tubing shall be plus or minus 10 percent, unless otherwise ndicated herein.

3.1.1 Polyethylene. Low density polyethylene, either sheet or extruded ubing, single or double wall shall be used for the outer pouch. As an

FSC 6530

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alternate, a laminate of paper and polyethylene shall be acceptable. The wall shall be 4 mils, minimum, total thickness each side of the pouch, and shall be capable of withstanding the 200 pound load test specified in 4.4.1 without rupturing or leaking.

3.1.2 Saran-mylar-polyethylene. Saran-coated transparent, 1/2 mil mylar laminated to 2 mils, medium density, transparent polyethylene sheeting shall be used for the inner pouch. Bursting strength of the laminate shall be between 80 and 90 pounds when tested as specified in 4.4.2. As an alternate, 2 mils, minimum, low density polyethylene tubing shall be acceptable provided the specified bursting strength requirement is met.

3.1.3 Chemical. Ammonium nitrate pellets or pellets of urea and potassium chloride shall be the chemical agent used in the cooling pad. The pellets shall be surfactant or organic coated to provide a timed release of the chemical into solution so that the performance requirements of 3.3 are met.

3.2 Construction. The cooling pad shall consist of one inner pouch filled with water and an outer pouch filled with chemical. The pouch materials and chemical shall be as specified in 3.1. The outer pouch shall be 5 to 6 inches wide by 7 to 10 inches long, exclusive of heat seals. The inner pouch may have a perforated break line or may be otherwise weakened to assist in breaking, and shall be of such size that it will fit inside the outer pouch while holding sufficient water to meet the requirements of 3.3. A water-soluble dark blue coloring dye may be mixed with the water before the inner pouch is sealed. A partial vacuum may exist in the outer pouch to assist in the rupturability of the inner pouch. It shall be possible for an adult person to activate the pad without a mechanical device or action, by squeezing the outer pouch against the inner pouch or by striking the pad with the fist in accordance with the manufacturer's instructions. The pouches shall be closed by means of heat seals which shall be straight, smooth and uniform in width along their entire length. The cooling pad shall meet the performance requirements in 3.3.

3.2.1 Marking. Each outer pad shall be permanently and legibly marked with activating instructions and a cautionary statement that, should there be accidental exposure of skin to the cooling solution, the affected area should be flushed with water. As an alternate, the above marking shall be on a securely attached label.

3.3 Performance. The surface of the cooling pad intended for contact with patient shall reach its minimum temperature, which shall be from 18°F. to 32°F., within 5 minutes of activation, and shall rise to no greater than 37°F. during the period 30 minutes after activation when tested as specified in 4.4.3. Leakage shall not occur after activation as described in 4.4.3.



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3.4 Workmanship. Workmanship shall be first class throughout. The ooling pad shall be free from defects which detract from their appearance r which may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the ontract, the contractor is responsible for the performance of all nspection requirements as specified herein. Except as otherwise speci- ied in the contract, the contractor may use his own or any ther facilities suitable for the performance of the inspection require- ents specified herein, unless disapproved by the Government. The overnment reserves the right to perform any of the inspections set forth n the specification where such inspections are deemed necessary to assure upplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for he contractor shall be maintained by the contractor and made available o the Government, upon the Government's request, at any time, or from ime to time, during the performance of the contract and for a period of years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined s both examination (such as visual and auditory investigation without the se of special laboratory appliances or procedures) and testing (determination y technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the anufacturers of the plastics and chemicals may be furnished in lieu of ctual performance of such testing by the contractor, provided lot identity as been maintained and can be demonstrated to the Government. The ertificate shall include the name of the contractor, the contract number, he name of the manufacturer or supplier, the NSN, the Item Identification, he name of the component/material, the lot number, the lot size, the ample size, the date of testing, the test method, individual test results, nd the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in ccordance with MIL-STD-105, and table I. Unit of product for sampling urposes is one individual pad.



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Table I. Sampling for examination

| | Inspection level | AQL (defects per hundred units) |
|---|---|---|
| For visual examination | | |
| Major defects | S-3 | 2.5 |
| Total defects (Major and Minor combined) | S-3 | 6.5 |
| For dimensional examination | S-1 | 2.5 |

4.2.2 For tests. Sampling for tests shall be in accordance with MIL-STD-105 and table II. Unit of product for sampling purposes is one individual pad. Acceptance level shall be zero regardless of lot size.

Table II. Sampling for tests

| Characteristic | Requirement | Test Procedure | Inspection Level |
|---|---|---|---|
| Physical properties | | | |
| Polyethylene | 3.1.1 | 4.4.1 | one test per lot |
| Saran-Mylar-polyethylene or plain polyethylene | 3.1.2 | 4.4.2 | one test per lot |
| Performance | 3.3 | 4.4.3 | S-3 |

4.3 Examination. The pads shall be examined to determine compliance with this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with classification of defects listed in table III. Examination shall not be restricted to the classified possible defects listed below.

Table III. Classification of defects

| Categories | Defects |
|---|---|
| Major | |
| 101 | Plastic pouches not free of holes, tears, cracks. |
| 102 | Pad already activated, and/or water in outer pouch. |
| 103 | Seams not completely heat sealed. |
| 104 | Inner pouch not filled with water. |
| 105 | Outer pouch has little or no chemical. |



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Table III. Classification of defects - continued.

| Categories | Defects |
|---|---|
| Minor | |
| 201 | Heat seals not straight, not smooth and uniform in width. |
| 202 | Plastic films not free of dirt, grease, loose foreign matter. |
| 203 | Directions for use and/or cautionary marking missing. |

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Outer pouch. A cooling pad complete with water-filled inner pouch and chemical-filled outer pouch placed on a plane surface shall be subjected to a 200 pound load gradually applied to an 8 by 10 inch plate placed on top of the pad.

4.4.2 Inner pouch. A water filled, heat sealed inner pouch, separate from the outer pouch, shall be subjected to a gradually applied load increasing from zero pounds until the pouch bursts. Test shall be performed as described in 4.4.1.

4.4.3 Performance. The cooling pad and a test blanket shall be conditioned at 72°F. ± 2°F. and 50% ± 10% R.H. for 24 hours prior to test. The test blanket shall be 100% wool and shall weigh 3 to 4 pounds. The blanket shall be folded into a 12 layer pack. The cooling pad shall be activated and vigorously agitated by shaking in a vertical plane for 30 seconds and immediately placed in the center of the folds of the blanket. A temperature sensor shall be placed at the center of the bottom surface of the cooling pad; pad resting on the sensor. The temperature shall be continuously recorded during the 30 minute test period using a suitable strip chart recorder. The temperature sensor shall be type T thermocouple, 30 AWG wire. Should multiple cooling pads be tested, a separate blanket pack shall be used for each and a separate channel shall be available on the recorder for each. As an alternate, an electronic thermometer with thermistor sensor (see 6.4) can be used for testing; however, the temperature must be continuously observed and recorded to detect the high and low temperature versus the time frame.

4.5 Examination of preparation for delivery. An examination shall be made to determine compliance with the packaging, packing and marking requirements of section 5. Defects shall be scored as specified in table IV. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with AQL of 4.0 expressed in terms of percent defective.



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Table IV. Examination of preparation for delivery

| Examine | Defect |
|---|---|
| Contents | not as specified. |
| Container | not as specified. |
| Marking | Omitted; incorrect; illegible; improper size, location, sequence or method of application. |
| Material | Component missing or damaged. |
| Workmanship | Bulging or distoration of containers. |

4.5.1 Examination of closure, waterproofing and banding of container. When the shipping container is required to comply with PPP-B-636, the examination for defects in closure, waterproofing or banding shall be in accordance with the appendix of the box specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Unit of issue. One package containing four cooling pads as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit package. Four cooling pads shall be packaged in a snug fitting corrugated fiberboard box of a suitable design, having a minimum bursting strength of 200 pounds. Closure of the box shall be adequate to prevent accidental opening.

5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Eight unit packages shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, type CF, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the appendix of the box specification.



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NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B. Eight unit packages shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, type CF, class domestic. Closure of the box shall be as specified in the box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Unit packages. In addition to any commercial markings, each unit package shall be marked in accordance with the requirements of the Federal Food, Drug and Cosmetic Act. Marking shall also include the National Stock Number, unit of issue, Item Identification, contract or purchase order number, level of packaging, and date packed.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Immediate first aid for injuries involving swelling, discoloration, pain and minor bleeding for which cold is recommended.

6.2 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 This specification covers the following item appearing in the Federal Supply Catalog:

| National Stock Number | Item Identification |
|---|---|
| 6530-00-133-4299 | PAD, COOLING, CHEMICAL, Plastic, Flexible Pack, Disposable, 4s |



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6.4 The Atkins Model 3L010 electronic thermometer with thermisto shall be suitable for use in the test specified in 4.4.3.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6530-1290
CS



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-P-37018 PAD, COOLING, CHEMICAL

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

**DD FORM 1 JAN 72 1426** REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

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